

**University of Victoria
Department of Electrical & Computer Engineering
ECE 299 – Introduction to Electrical and Computer Engineering Design
Final Report
Summer Term ECE 2026**

Clock Radio Project

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Executive Summary

Electrical and Computer Engineering 299 is an introductory course to engineering design within the scope of Electrical and Computer Engineering. The Clock Radio project intends to familiarize students with critical aspects of design and development of an electronic product. Students become proficient with electrical circuits, PCB design, and professional practices for the development process of a product. The Clock Radio (designed by students in groups of two) must operate on a Raspberry Pico 2W and be capable of the following [1];

- Set and Edit time through a user interface
- Set and Edit alarm time through a user interface
- Trigger alarm at the set alarm time
- Snooze and turn off alarm through a user interface
- Tune FM radio frequency through a user interface
- Play radio output to a speaker
- Volume control through a user interface
- Display radio channel to the user

Additionally, students will have to design a two-layer PCB for the project and have it manufactured, solder components onto it, followed by designing and 3d printing an animal-themed enclosure to house the PCB and other components. The final part of the project entails demonstrating the final project and writing a final report. See figure 1 for the image of the final project.



Figure 1-Final Radio Clock

Table of Contents

Table of Contents

Executive Summary	1
Table of Contents	3
List of Figures	4
List of Tables	4
Project Goal.....	5
Constraints.....	5
Project Expectations.....	6
Design Choices.....	7
Project Design Choices	7
PCB.....	7
Soldering.....	7
Enclosure Design Choice	7
Final Report/Project Demonstration	8
Clock Design Choices	8
Set/Edit Time and Alarm	8
Snooze/Dismiss Alarm	8
Select Field and Adjust Value	8
Radio Design Choices	8
Tune Radio/Adjust Volume	8
Radio output	8
Project Planning	9
Circuit Schematic & PCB Design	13
Bill of Material and Cost Analysis	16
Enclosure Design.....	17
Testing and Validation	18
Test Procedure 1 – Hardware Testing	18
Test Procedure 2 – PCB Testing	19
Steps Taken in Resolving Issue:	19
Test Procedure 3 – 3D Enclosure Testing.....	19

Test Procedure 4 – Smaller Testing & Validations	20
Display Module Test	20
Encoder Module Test	20
FM Radio & Amplifier Module	21
Code of Ethics	21
Conclusions & Recommendations	22
References	23
Appendix	24
Checklist	25

List of Figures

Figure 1-Final Radio Clock	2
Figure 2 - Prototype	11
Figure 3 - Gantt Chart	12
Figure 4– Design Schematic	13
Figure 5- PCB Design	14
Figure 6- 3D View of PCB	15
Figure 7– Front /Back of PCB.....	15
Figure 8– 3D Enclosure of Project.....	17
Figure 9 – Sketches of 3D Enclosure	18
Figure 10, 11 – Tested Enclosures.....	20

List of Tables

Table 1– Roles of team members	9
Table 2- Task of team member #1	10
Table 3-Tasks of team members #2	10
Table 4– Cost of Components in Project	16

Project Goal

Teams of two are expected to design a PCB to house all the components necessary to create a working radio and alarm clock, along with designing and 3d printing an enclosure to house the PCB and other components. Through skills and tools learned in ECE299, students must ensure that the project adheres to the functionality specified in the executive summary.

Constraints

The design of the clock radio was subject to several constraints specified by the project requirements. Below is a list of constraints in which the project was subject to [1]:

- **Microcontroller:** The system must utilize a Raspberry Pi Pico 2W as the central processing unit to orchestrate all core processing, concurrent tasks, and peripherals.
- **Programming Language:** All system firmware must be developed in MicroPython, with the optional integration of Claude Code to assist in software optimization and debugging.
- **Communication Protocol:** The OLED display must interface with the microcontroller via the Serial Peripheral Interface (SPI).
- **PCB Requirements:** The design must be implemented on a two-layer PCB and have all components soldered onto it.
- **Team Requirements:** The project must be designed, built, and demonstrated by a collaborative partnership of two students registered within the same laboratory section.
- **3D Development:** A fully functional prototype housed in a custom 3D-printed, animal-themed enclosure must be completed and ready for evaluation by the final week of labs.
- **Financial Responsibilities:** All costs for components, custom PCB fabrication, and structural materials are the financial responsibility of the team members.

Project Expectations

The expectation of this project is to design, implement, and validate a functional, animal-themed clock-radio prototype that incorporates the following system-level functionalities [1].

- **Design a two-layer PCB (Printed Circuit Board):** To integrate all electrical components and coordinate its production with a commercial fabrication company.
- **Solder:** All electronic components must be soldered on to the manufactured PCB (Printed circuit Board).
- **Design and Print an Animal-Themed Enclosure:** The PCB and power supply must be all enclosed into an animal themed 3D printed enclosure.
- **Final Report / Project Demonstration:** There must be a final report submit before July 21, and a project demonstration must be presented before the last week of the labs.

Clock Functionalities:

- **Set/Edit Time:** Configure system time parameters directly via interactive OLED menu.
- **Set/Edit Alarm:** Configure alarm time, storing the values within the system memory (interactive OLED menu).
- **Snooze/Dismiss Alarm:** Manage active alarm states using physical push-button hardware.
- **Select Field:** Click the hardware button to cycle through editable values (Time Hours/Minutes, Alarm Hours/Minutes).
- **Adjust Value:** Rotate the encoder shaft to increment or decrement the active configuration parameter.

Radio Functionalities:

- **Tune Radio:** The radio clock must at least tune to one FM radio channel.
- **Radio Output:** The radio audio must output, out of a speaker.
- **Volume Control:** Audio level must be control with input/output hardware devices.
- **Display Radio:** Radio Info displayed must be display on an OLED screen.

Design Choices

The following section outlines the major design decisions made during the development of the clock radio system. Each design choice is supported by engineering principles, component specifications, datasheets, and design calculations to justify the selection of hardware, software architecture, and mechanical implementation.

Project Design Choices

Below, are design choices adhering to the project specifications themselves, not including design choices for the clock and radio.

PCB

The team chose to fit all components onto a 2-layer PCB of size 55.0mm x 54.5mm to fit within the 30 cm^2 size constraint [1]. The team chose to design the PCB (and schematic) in KiCad for ease of use. Every component is soldered onto the PCB via through holes. Several components (Raspberry Pico 2W, rotary encoder, 2.42 OLED screen, radio module) were mounted onto pin headers, rather than directly onto the board. Two buttons and the speakers are connected to the PCB via terminal blocks. The choice to use pin headers and terminal blocks was made so it was easy to remove components for debugging purposes, and for re-use of components in the future without having to desolder them entirely.

Soldering

All components were hand soldered at home using a student's personal soldering setup/equipment. The soldering process entails adding flux to the component being mounted, press the hot soldering iron to the pin and through-hole, apply solder to through-hole, remove solder, and remove soldering iron to allow solder to harden.

Enclosure Design Choice

The team chose to design the enclosure in Fusion 360 using a free educational license. The enclosure was designed to fit all electrical components on the inside, and have the buttons, screen, encoder, and speaker all mounted on the outside with no wires showing. The edges of the enclosure are rounded to 4mm radius to ensure no sharp edges. The enclosure can be opened to access the inside and is held together by magnets, so it is easy to open and close. To fit the design constraint of an animal theme [1], the team 3d-printed a rooster head to mount on the lid of the enclosure. When 3d printing the enclosure, the team chose to print the enclosure in four pieces to minimize overhang (to waste less filament used in supports) and assemble after it was printed.

Final Report/Project Demonstration

The final report is to be submitted by the team on July 21st, 2026, and the demonstration is held during the last week of labs on July 23rd, 2026, as per the project expectations on the ECE 299 webpage [3]

Clock Design Choices

The next section states the design choices that were made for the clock functionality of the project.

Set/Edit Time and Alarm

The team implemented editing the time and alarm with the rotary encoder. A push button cycles between several modes, including but not limited to clock hour, clock minute, alarm hour, and alarm minute. Turning the encode in the selected mode adjusts the indicated value. Alarm time, and clock time are displayed on the OLED screen connected via I2C.

Snooze/Dismiss Alarm

A push button was used to implement this feature. While the alarm is going off, the button can be pressed quickly to snooze, and a long press stops the alarm entirely. The alarm feature can also be completely disabled via a long press of the same button (provided the alarm is not already sounding). The status of the alarm is displayed on the OLED screen.

Select Field and Adjust Value

As described above, selecting a field is done with the push of a button, and adjusting the value is done with the rotary encoder.

Radio Design Choices

This section describes the design choices made with respect to the radio expectations.

Tune Radio/Adjust Volume

Radio tuning is done via the rotary encoder. There is a minimum and maximum frequency, so when the minimum or maximum frequency is reached, it can cycle back to the start or end. Adjusting volume is also done using the rotary encoder. The volume has range from 0 up to 15. Frequency and volume adjustment are both modes that can be cycled to using the select field pushbutton. This is displayed on the OLED screen.

Radio output

The radio output goes to the LM386 amplifier, then to the speaker. Communication to the radio module is done using the I2C protocol.

Project Planning

Before presenting the design process, our team assigned roles and responsibilities to ensure that all aspects of the project were completed efficiently. Each member was responsible for specific tasks based on their skills and interests while also collaborating throughout the project when needed. Table 1 below summarizes the roles and responsibilities assigned to each team member.

Table 1– Roles of team members

Member	Hardware	Software
Team member 1 Colton Restell	• Design Radio Speaker • Rotary Encoder • Debugging	• Display (Volume, Radio station etc.) • Polling Radio information (frequency) • Clock System
Team member 2 Cole Alexander	• SPI protocol for Display • Rotary Encoder • LM386 Amplifier Wiring	• Code associated with rotary encoder (press to switch between volume/frequency, turn to adjust selected option) • Debugging • Alarm System

Next, we developed a project timeline to organize the major milestones and ensure that the project remained on schedule. This timeline helped the team track progress, manage deadlines, and coordinate tasks throughout the development process. The complete project schedule is shown in the table below. See Table 3 and 4.

Team member 1: Colton Restell

Table 2- Task of team member #1

Task	How will you test successful completion?	Date of delivery
Acquire frequency from radio module to display on screen	Verify the correct frequency shows up on the screen	June 21
Rotary encoder adjustment values adjust live on screen	Demonstrate volume level and frequency updating on-screen when turning the encoder	June 28
Code Clock	Ensure time displays and can be adjusted, and stays in sync with real clock	July 7
PCB Design of screen and encoder section	Test functionality of PCB upon arrival	July 18

Team member 2: Cole Alexander

Table 3-Tasks of team members #2

Task	How will you test successful completion?	Date of delivery
Program Rotary output	Ensure rotary encoder value updates itself when changed, and that a button press on the encoder is registered	June 28
Integrate rotary encoder	Verify tuning and volume control work correctly	July 12
Code Alarm	Ensure alarm goes off at set time, and alarm time can be adjusted or disabled completely	July 7

Software debug	Ensure code works flawlessly	July 18
3D Enclosure	Design an enclosure that PCB and other components	July 20

After assigning roles and responsibilities, then the team members decided on an audio transmission for the speaker. Hardware and firmware began simultaneously, a breadboard (figure 2) prototype was assembled to test the Raspberry PI, OLED display, audio amplifier, speaker and output/input devices. At the same time, the software was developed to implement the user interface and logic for the circuit. A GitHub repository was created allowing seamless merging of code. This approach allowed us to test both hardware and software at the same time allowing us to catch errors early. After validating the prototype, using a KiCad a custom printed circuit board (PCB) was designed to replace the prototype breadboard.

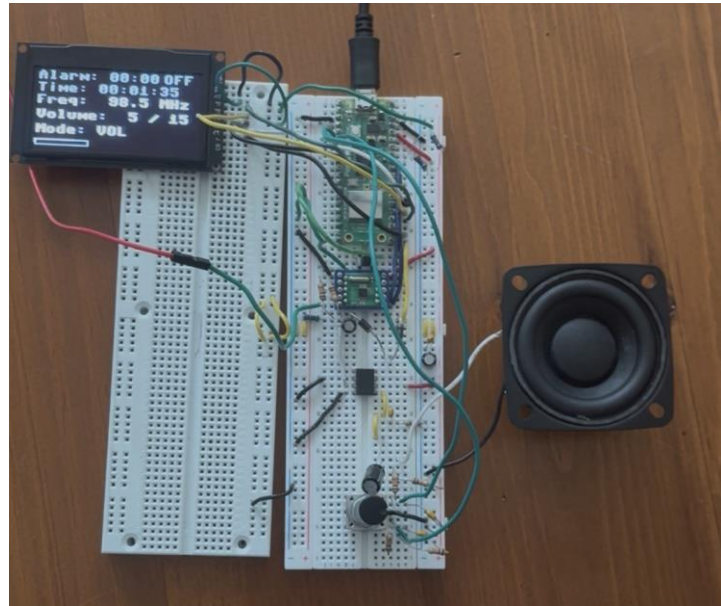


Figure 2 - Prototype

Then, in Fusion360, a custom enclosure was designed to house the components. The enclosure was designed to provide structural support, protect the internal electronics, and create a clean professional appearance while ensuring easy access to the user interface and power supply. All while following the constraint of needing to be animal themed.

A Gantt Chart (figure 3) outlining the full production of the radio clock is demonstrated below.

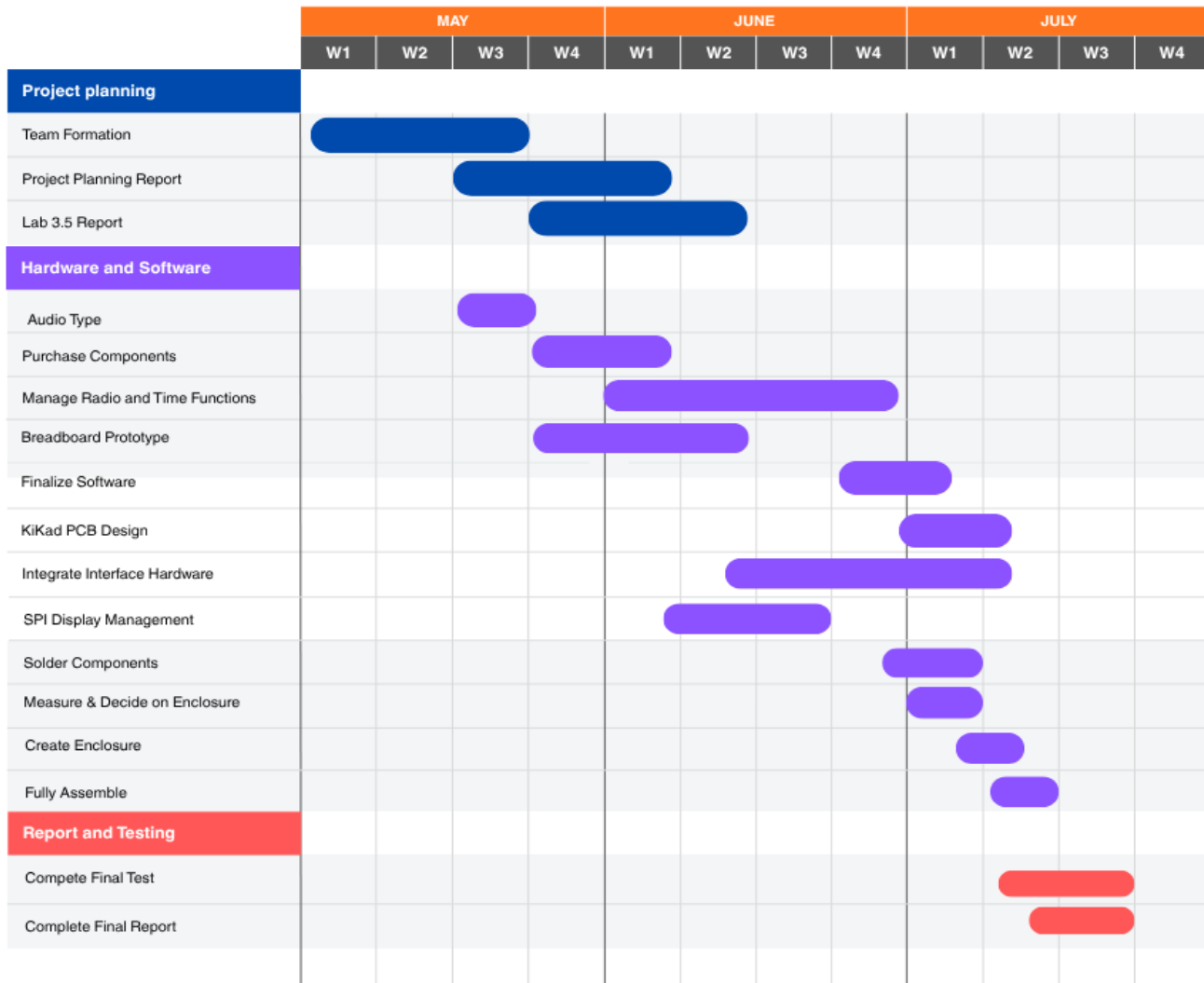


Figure 3 - Gantt Chart

Circuit Schematic & PCB Design

This section outlines the specifics regarding the schematic and PCB design of the project. Below, is an image of the schematic (figure 4).

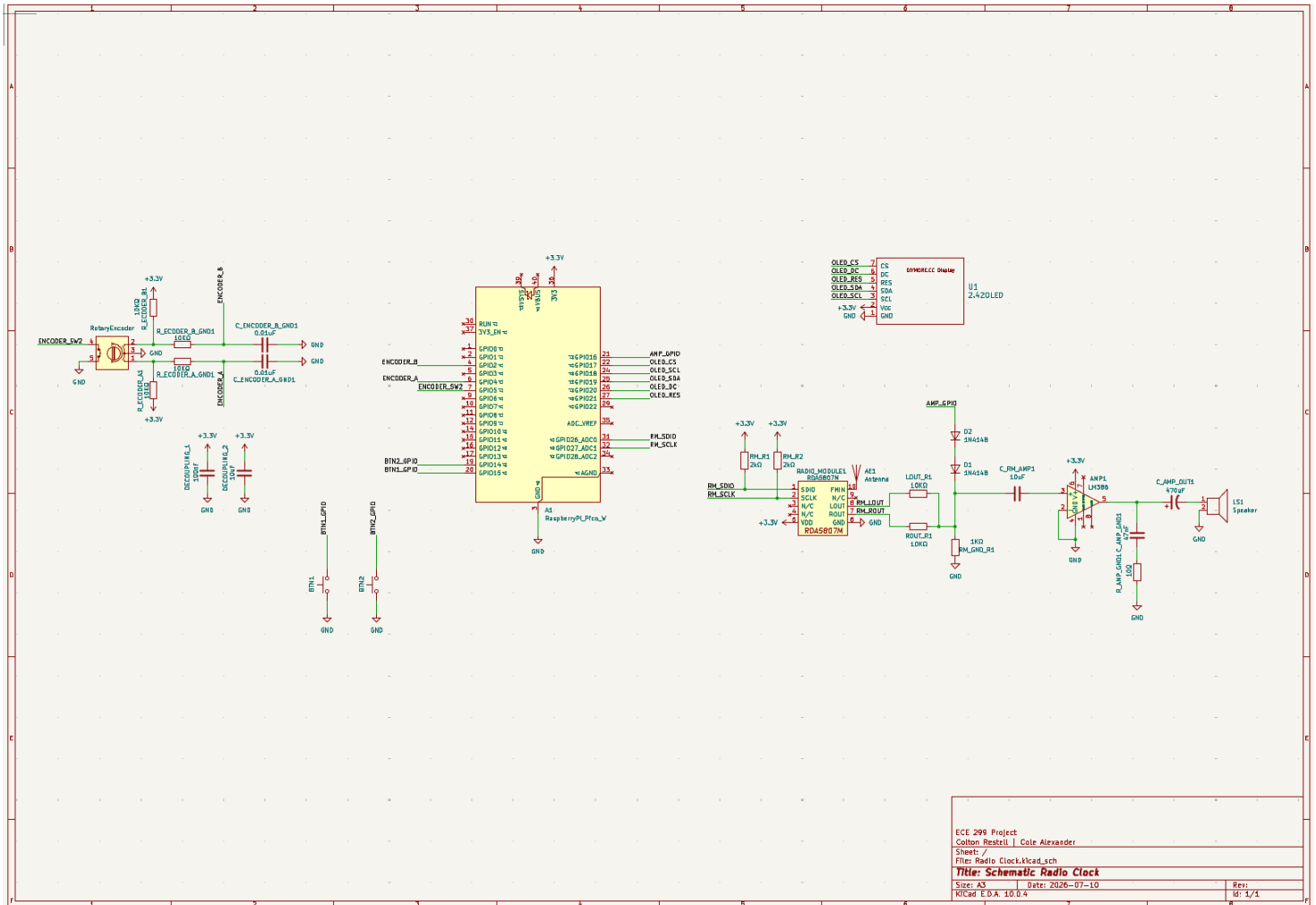


Figure 4– Design Schematic

This schematic shows the overall circuit design of our radio clock system. The design was divided into several functional blocks: the microcontroller, display interface, radio module, user input controls, decoupling capacitors, etc. See figure 5 for the PCB design.

- **Microcontroller (Raspberry Pi Pico 2W)**
 - Provides enough GPIO pins and communication protocols required for the display, radio module and user input

- Internal regulator on Pico supplies 3.3 Volts to all components with VCC/VDD
- **OLED Display (SSD1309)**
 - Uses SPI communication protocols, providing high data transfer rates then the other I^2C 0.96 INCH OLED screen.
 - SSD1309 OLED Display (128 x 64 Pixels) Connected SPI
 - SCL--> GPIO 18
 - SDA --> GPIO 19
 - RES --> GPIO 21
 - DC --> GPIO 20
 - CS --> GPIO 17
- **Radio Module**
 - Uses I2C, module only requires two-line communication, reducing complexity and fits into the project constraints.
 - SDIO--> GPIO 26
 - SDA --> GPIO 27
- **Rotary Encoder / Buttons**
 - Included to allow user interaction with the clock. The rotary encoder provides adjustable input for settings such as time and volume, while buttons provide additional control functions.
 - Buttons are pulled to HIGH by default and low when pressed
 - Encoder:
 - A --> GPIO 4
 - B --> GPIO 2
 - Increments selection by ± 1
- **Pull-up Resistors**
 - Included in circuit to establish a default HIGH logic level on input pins, when buttons or switches are not pressed. When the button is activated, the input is connected to ground, creating a LOW signal
- **Capacitors**
 - Used to reduce electrical noise and stabilize the supply voltage, improving the reliability of the circuit.

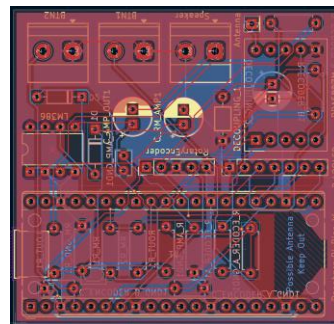


Figure 5- PCB Design

All the above, were features that were taken into consideration while designing the PCB, ensuring appropriate GPIO pin assignment, power distribution, and component placement to support, SPI, I^2C and hardware user input and output. Below are images of the PCB (see figure 6).

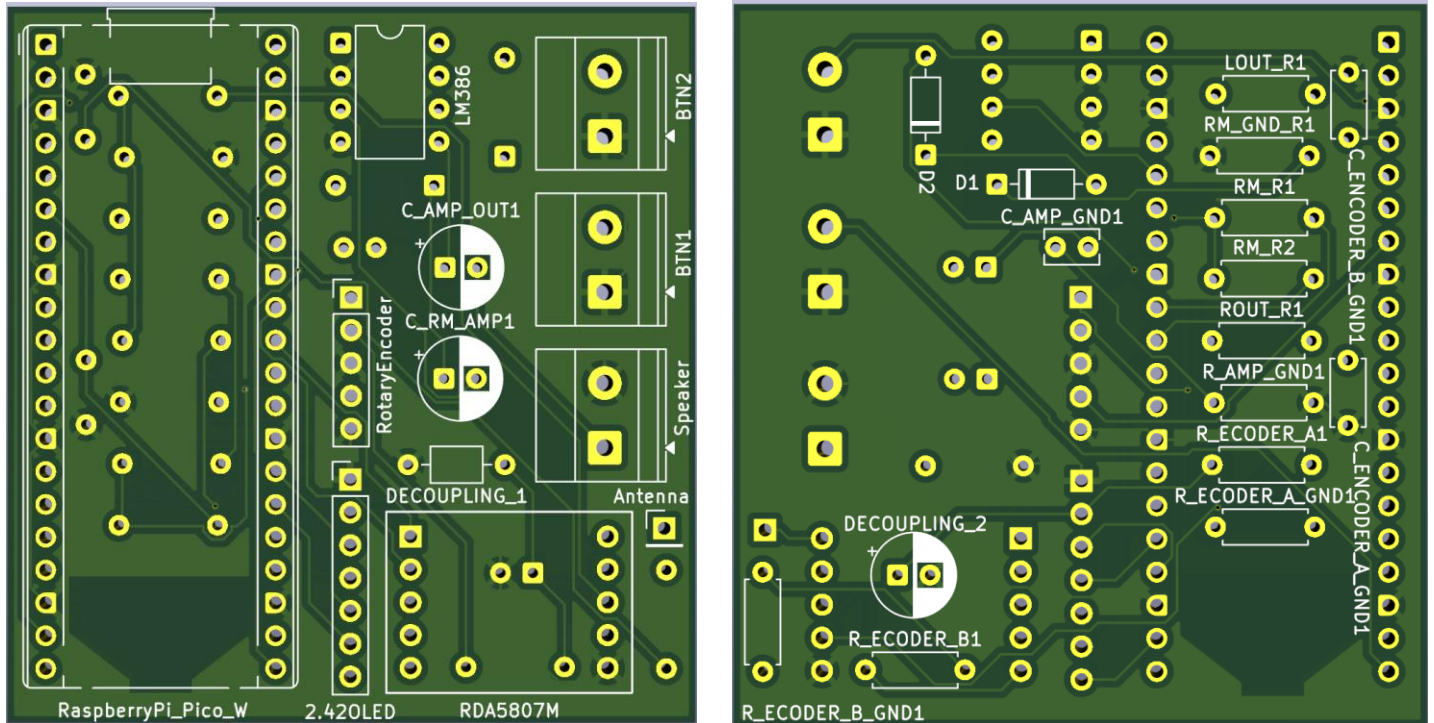


Figure 6- 3D View of PCB

This PCB was designed to integrate all components of the radio clock into a compact and reliable circuit that was 30 cm^2 in area to ensure compatibility with the enclosure. A key design consideration was preserving the clearance zone around the Raspberry Pi Pico 2W's internal antenna; no traces or copper pours were routed in this region to prevent interference with wireless communication. The PCB connects the microcontroller, OLED display, radio module and user input components while providing proper electrical connections between the components. Finally, to reduce wiring complexity and improve assembly and debugging, all external components were placed strategically. Some extra precautions were taken into the PCB layout (see figure 7), such as adding an extra ground and power pin for debugging and an extra button in case of addition of features in the software.

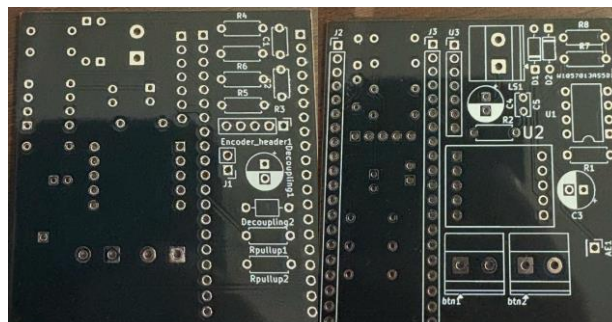


Figure 7- Front /Back of PCB

Bill of Material and Cost Analysis

The table below shows the cost of each item we used in our final project along with the part name, source of cost information and description (see table 4).

Table 4– Cost of Components in Project

Label in Schematic	Component Description	Part Number	Cost/Unit Quantity	Source of Cost Information
A1	Raspberry PI Pico 2W	N/A	\$9.80	PiShop
U1	OLED Display	N/A	\$13.00	ECE Store
RotaryEncoder	Rotary Encoder	PEC11R	\$2.00	ECE Store
Radio_Module1	Radio Module	RDA5807M	\$2.00	ECE Store
Amp1	Audio Amplifier	LM386N-4	\$1.00	ECE Store
BTN1 & BTN2	Switch Button	N/A	\$0.25	ECE Store
N/A	PCB	N/A	\$5.00 USD	PCBWay
N/A	Power supply	N/A	\$12.50	Canadian Tire
LS1	Speaker	ABS-216-RC	\$5.00	ECE Store
LOUT_R1 ROUT_R2 R_Encoder_A1 R_Encoder_B1 R_Encoder_A_GND1 R_Encoder_B_GND1	10K Oms Resistor	N/A	Free	ECE Store
RM_R1 RM_R2	2k Oms Resistor	N/A	Free	ECE Store
RM_GND_R1	1k Oms Resistor	N/A	Free	ECE Store
R_AMP_GND	10 Oms Resistor	N/A	Free	ECE Store
C_AMP_OUT1	470uF Capacitor	N/A	Free	ECE Store
C_AMP_GND1	47nF Capacitor	N/A	Free	ECE Store
C_RM_AMP1 Decoupling_2	10uF Capacitor	N/A	Free	ECE Store

D1 / D2	Diode	1N4148	Free	ECE Store
Decoupling_1	100nF Capacitor	N/A	Free	ECE Store
C_Encoder_B_GND1 C_Encoder_A_GND1	0.01uF Capacitor	N/A	Free	ECE Store
N/A	20 Socket Header	N/A	\$1.50 x2	ECE Store
N/A	7 Pin Header	N/A	\$0.5	ECE Store
N/A	5 Pin Socket	N/A	\$0.5 x2	ECE Store
N/A	Terminal Block	N/A	\$0.25 x3	ECE Store

Enclosure Design

This section explains the enclosure design process. The enclosure was designed with the intent to house all internal components as compact as possible, while concealing all electrical components from the outside view. Below, are 3d views of the enclosure itself (figure 8).

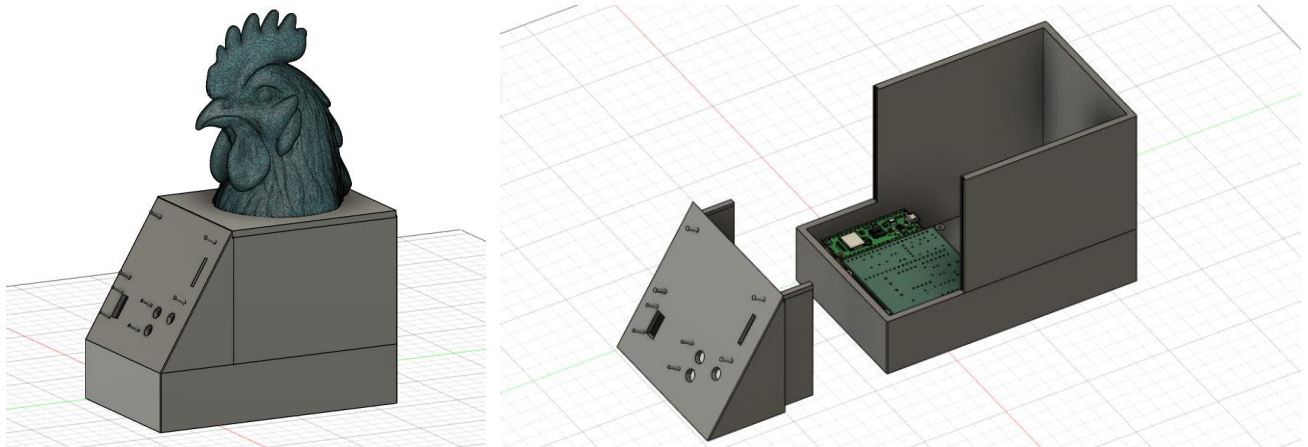


Figure 8– 3D Enclosure of Project

To fit the animal theme, the team added a 3d printed rooster head to the lid of the enclosure. Explained in more detail later, we added multiple alarm sounds to choose from, one of which is a rooster noise that one may hear in the morning. Below are the drawings associated with the enclosure (figure 9).

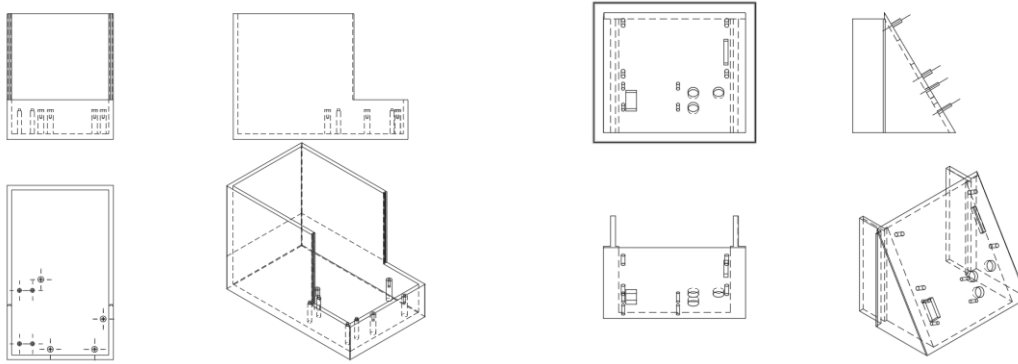


Figure 9 – Sketches of 3D Enclosure

The slanted piece on the front was made to hold the screen, speaker, buttons and encoder. The team used panel mount buttons and rotary encoder to screw through their respective holes. The speaker and screen mounts onto four pins each, with a hole for cables in the back. Inside the box, the team integrated vertical cylinders which would hold the Raspberry Pico in place, and standoffs into the 3d print to screw self-tapping screws into which would hold the PCB. The standoffs did not work as expected (see “Testing and Validation,” for details) and we instead opted with a sliding mechanism with grooves that the edges of the PCB would slide into. The lid and the slanted pieces both mount to the box using magnets, so they are easy to remove to access the internal components.

Testing and Validation

This section outlines the testing and validation processes taken to ensure the radio clock functions as described.

Test Procedure 1 – Hardware Testing

Throughout the assembly of the radio clock, there was multiple times where the button inputs wouldn’t register. To overcome and troubleshoot this, our team added print statements to determine where the source of the problem was coming from. This problem also occurred with our encoder. The team also had issues getting the display to work consistently, so team members decided to test the screen functionality on its own as well. See to the GitHub repository linked in the appendix for test code to validate button, encoder, and display inputs.

The testing results proved that the system Raspberry Pi Pico 2W was able to read the inputs from the GPIO pins, from this we determined that the root of the issue occurred in our logic of the code.

Test Procedure 2 – PCB Testing

During PCB testing, a design issue was discovered with the custom Raspberry Pi Pico footprint. The footprint was created by hand initially had an error in the pin placement which led to the Pico being connected incorrectly on the PCB. The schematic was updated to fix the footprint, but the error was discovered after the PCB was manufactured and there was not enough time to have a new PCB manufactured. To keep the project on schedule, testing proceeded by re-routing the affected connections on the PCB with jumper wires. It was not the best solution from a manufacturing perspective, but it did restore the correct electrical connections and the board worked as it should.

Steps Taken in Resolving Issue:

I. Initial Power Test:

- Using a multimeter, we ensured that voltage drop between each component was as expected.

II. Pin/Footprint Verification:

- Compared the custom footprint with the physical Raspberry Pi Pico 2W (Datasheet). This confirmed our problem and one of the rows of the Raspberry Pi Pico 2W was rotated by 180 degrees on the PCB.
 - Using jumper wires and rerouting the Raspberry Pi Pico 2W to the header resolved the issue.

Test Procedure 3 – 3D Enclosure Testing

Due to the changes on the PCB, the initial 3D printed enclosure no longer fit the Raspberry Pi Pico, as the extra jumper wires required more space around the microcontroller. In addition, the PCB design did not have mounting holes and therefore the board could not be fixed with screws.

To solve these problems, some enclosure prototypes were designed and 3D printed. Each revision was verified for component clearance, wire routing, and PCB stability. The design was modified between prints to provide more space for the rewired microcontroller, and to introduce alternative features that held the PCB securely without screws. A final enclosure was produced after several iterations that fit all the components properly and securely supported the PCB while allowing reliable operation of the finished radio clock. The first image below, figure 10, is the first iteration of the mounting bracket for the PCB. Due to measurement error, the bracket was too wide, and the PCB did not slide in. The second

image, see figure 11, is the printed part intended to mount the screens, speaker, buttons and encoder onto. There were two issues with that. The first was that the pins for the speaker were not long enough, so the speaker could not mount. The second issue was that the holes for the buttons were too close together. Both issues led to us having to re-print those components.

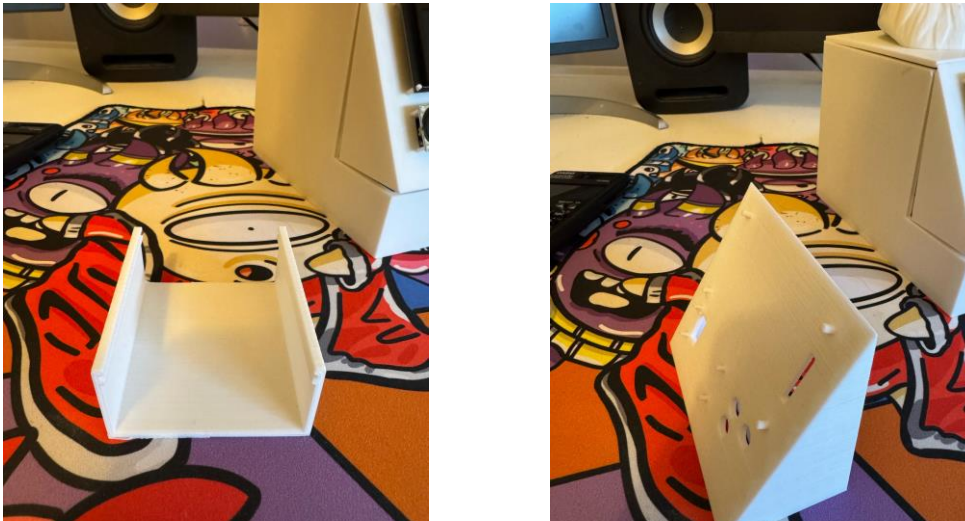


Figure 10, 11 – Tested Enclosures

Test Procedure 4 – Smaller Testing & Validations

Display Module Test

While testing the display using the functions from code supplied in the lab **‘basicdisplay.py’** [2] from this file, it displayed an error. This error was due to us purchasing a larger screen which thus required a different driver. Using a file on a GitHub repository we found a valid driver that was compatible with our OLED screen.

Encoder Module Test

The encoder functionality was tested and validated using the **‘encoder.py’** [2] code provided in the lab exercises. Initial testing was performed on the breadboard prototype to verify correct operation prior to integrating into the final PCB design. Hardware debouncing was incorporated to minimize the impact of contact bounce, ensuring a single, stable input for each encoder rotation and avoiding false or multiple counts.

FM Radio & Amplifier Module

Tests were performed for the audio amplifier and the FM radio module supplied in the lab [2]. The radio module was tested and verified to successfully receive FM stations and output audio through the speaker. The only problem encountered in testing was poor radio reception because there was no antenna. The answer was to attach a long wire to the antenna input of the radio module, and this made for a much better signal and the radio was able to better pick up stations.

Code of Ethics

The relevant list of ethics pertaining to this project include the following from the EGBC “Code of Ethics” [3]:

- I. Hold paramount the safety, health, and welfare of the public, including the protection of the environment and the promotion of health and safety in the workplace.**
 - Our team made health and safety a priority, especially when dealing with soldering. Many precautions were taken such as soldering in an open environment with a window open and using correct equipment. Additionally, ensuring that our workspace was clean and have an extra pair of free hands when needed.
- I. Practice only in those fields where training and ability make the registrant professionally competent.**
 - Both team members took required training through UVic Maker Space, and both members also have prior experience with soldering and 3D printing. By having these perquisites, we were competent in the design stage of the radio clock.
- I. Maintain competence in relevant specializations, including advances in the regulated practice and relevant science.**
 - Both member of the team stayed up to date with the lecture material and lab sections, such that both members were informed of the constraints of the project.

I. Conduct themselves with fairness, courtesy, and good faith towards clients, colleagues, and others, give credit where it is due and accept, as well as give, honest and fair professional comment.

- Collaboration was crucial to our project. All team members contributed equally, and credit was given accordingly. Feedback was exchanged in a respectful manner, and constructive criticism was received in the spirit of improvement.

Conclusions & Recommendations

Over the course of this project, the team was successful in building a clock radio. While doing so gained valuable hands-on experience and developed critical engineering skills through testing, problem solving and collaboration. The team also became more proficient in soldering, KiCad, Fushion360, and communication protocols.

Despite the team's achievements, we encored several challenges that tested our problem solving and troubleshooting skills. These challenges provided valuable learning experiences and required the team to systematically identify issues, evaluate possible causes, and implement effective solutions. Critical issues include physical hardware failure such as non-responsive power supply, enclosure misalignments due to measurement inaccuracies and modify hardware components.

In conclusion, our design satisfied all the course constraints and criteria. A Raspberry Pi Pico 2W was used to drive the radio clock and was programmed with MicroPython to control the functions of the system. All components were held securely in a custom designed PCB assembled in a 3D printed case to create a final product.

After completing this project, we have complied the following recommendations based on experiences and observations:

- Design the PCB with mounting holes to simplify enclosure integration and assembly.
- Test different antenna designs and configurations to improve radio signal strength and reception reliability.
- Consider a larger enclosure to allow more flexibility for component changes.
- Prototype the 3D-printed case earlier to reduce redesigns caused by hardware changes.
- Add additional filtering components to reduce electrical noise affecting sensitive modules.

References

- [1] “ECE 299 Project Expectations.” Accessed: July 21, 2026. [Online]. Available: [Project expectations and marking scheme - Summer 2026 ECE 299 A01 \(30299\)](#)
- [2] “ECE 299 Lab Manual.” Accessed: July 20, 2025. [Online]. Available: [ECE299 Lab Manual - updated on June 12 - Summer 2026 ECE 299 A01 \(30299\)](#)
- [3] “Code of Ethics.” Accessed: July 21, 2025. [Online]. Available: <https://www.egbc.ca/Complaints-Discipline/Code-of-Ethics/Code-of-Ethics>

Appendix

To view our radio clock, visit our GitHub repository:

<https://github.com/ColtonRestell/Radio-Clock.git>

Checklist

ECE 299 Report: Checklist

Format:

1. **Body-text** uses consistent font type and size
YES ☒ NO ☐ NOT SURE ☐
2. **Titles** and **Headings** are easy to distinguish from the body text
YES ☒ NO ☐ NOT SURE ☐
3. **Pages** and **equations** are numbered
YES ☒ NO ☐ NOT SURE ☐
4. **Symbols** used in equations are defined the first time you use them
YES ☒ NO ☐ NOT SURE ☐
5. Information in paragraphs has been made easy to read whenever possible by way of lists, figures, tables etc.
YES ☒ NO ☐ NOT SURE ☐
6. All information derived from external sources is properly and fully cited in-text, using IEEE citation style
YES ☒ NO ☐ NOT SURE ☐
7. A properly formatted **IEEE-style References list** closes the document:
YES ☒ NO ☐ NOT SURE ☐

Figures and Tables:

1. Whenever possible, images are original, not directly copied from a source
YES ☒ NO ☐ NOT SURE ☐
2. Figures and Tables are referred to in text (e.g. See fig 1; see Table 2, etc.)

before they appear on the page

YES ☒ NO ☐ NOT SURE ☐

3. Each image and table includes ***sequentially numbered captions***

(in Times New Roman font, size 10) that explain what the image or table is about

YES ☒ NO ☐ NOT SURE ☐

4. **Graphs** provide clearly labeled **axes**, with accompanying **legend**.

YES ☒ NO ☐ NOT SURE ☐

5. **Images** are completely readable without magnification (i.e. zooming in).

YES ☒ NO ☐ NOT SURE ☐

Style

1. Consistently uses either **first person** or **third person** narration.

YES ☒ NO ☐ NOT SURE ☐

2. Supplies **written descriptions** of figures and tables to clarify their relevance to the reader.

YES ☒ NO ☐ NOT SURE ☐

3. Makes **direct reference** to each figure and table in the presented description (e.g. *See Fig. 1; as shown in Table 2*, and so forth).

YES ☒ NO ☐ NOT SURE ☐

4. Includes **signal phrases** (e.g. *in addition; however; in contrast; as a result*, etc.) to clearly map the logic of the report.

YES ☒ NO ☐ NOT SURE ☐

5. Maintains a **formal**, though not “**stuffy**” or “**inflated**” tone; it avoids informal **contractions** (*won't; can't*; etc.).

YES ☒ NO ☐ NOT SURE ☐

6. Uses **clear active sentences**, i.e. state the main topic or **subject** at the beginning of the sentence, closely followed by an **action word**—a verb.

YES ☒ NO ☐ NOT SURE ☐

7. Maintains **agreement between subjects and verbs** (i.e. a singular/plural sentence topic is followed by a singular/plural verb).

YES ☒ NO ☐ NOT SURE ☐

8. Uses **simple words** when simple words will do the job, following the **plain language** principles.

YES ☒ NO ☐ NOT SURE ☐

9. Eliminates **phrases** (e.g. *in view of the fact that*) when a **single word** (e.g. *because*) would do instead.

YES ☒ NO ☐ NOT SURE ☐

10. Links **pronouns** (stand-in words like *this*, *they*, or *it*), back to a specific noun (i.e. an object, concept, or person) in the previous phrase or sentence.

YES ☒ NO ☐ NOT SURE ☐

11. Supplies **grammatically parallel** lists and headings, meaning each item in a list and each heading takes the **same grammatical form** (e.g. all verb phrases or all noun phrases).

YES ☒ NO ☐ NOT SURE ☐

12. Has been carefully proofread to ensure **punctuation** (commas, colons, semi-colons, and periods) aligns with Standard English conventions to enhance accuracy and clarity.

YES ☒ NO ☐ NOT SURE ☐

REFLECTION: what improvements did you make to your report because of completing this checklist? Identify 2 specific changes.

After completing the checklist, several improvements to the report were made to improve clarity. First, all the format of the report was revised, include tables, figures, citations and overall format of document. Second, the structure and wording were revised, improving readability and consistency of the document.